

Brain Tumor Classification using Transfer Learning and Grad-CAM

BRISC 2025 Dataset (Classification Task)

🎓 Tafila Technical University · Faculty of Information & Communications Technology

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94.8%

Accuracy

3

Models

Introduction & Medical Motivation



Manual Diagnosis Challenges

Time-Consuming

Manual interpretation takes hours per scan

Expertise Variability

Depends heavily on radiologist skill

Workload Increase

Growing MRI scan volume overwhelms staff

Automated Solution Benefits

Faster Analysis

Immediate results, no waiting time

Consistent Quality

Reduced variability in diagnoses

Support Tool

Assist radiologists, reduce errors

Problem Statement & Objectives



Problem Statement



Manual brain tumor diagnosis is **time-consuming** and prone to **expertise variability**

Increasing MRI scan volumes overwhelm radiologists, leading to delayed decisions and inconsistent diagnoses. An automated system is needed for fast, consistent, and accurate classification.



Main Aim

Develop **automated system** for brain tumor classification using **deep learning** and **transfer learning**



Objectives

01

Analyze brain MRI images using CNN-based models

02

Implement baseline CNN for reference benchmarking

03

Apply transfer learning with pretrained models

04

Evaluate and compare model performance

05

Identify strengths and limitations of each approach

Dataset Description – BRISC 2025



5,000

Training Images
15% validation



1,000

Test Images
Final evaluation



224×224

Image Size
Standardized input



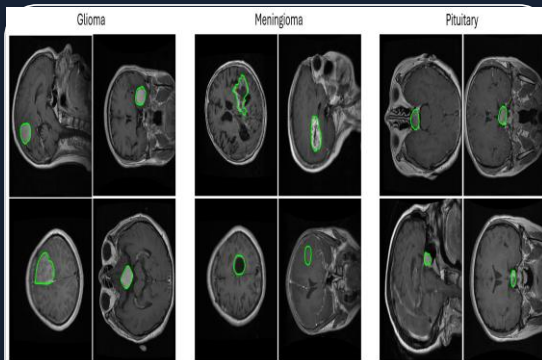
4

Classes
Multi-class task



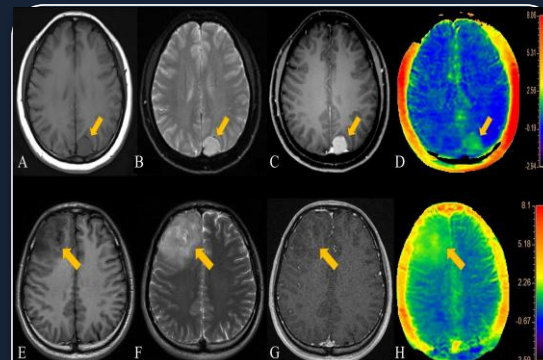
Real Medical Cases

BRISC 2025 Dataset



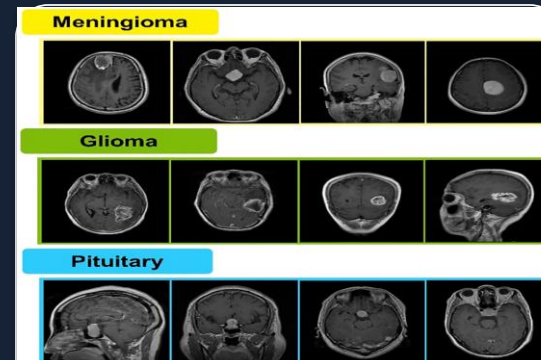
Glioma

Most common brain tumor type



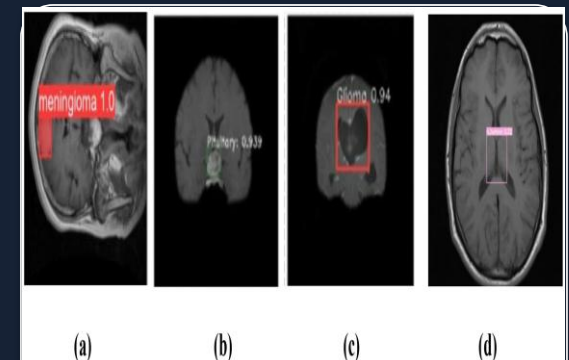
Meningioma

Arising from meninges



Pituitary

Benign tumor of pituitary gland



No Tumor

Healthy brain MRI



Class imbalance reflects realistic clinical distribution

Handled through class weighting during training to ensure fair representation across all categories



Important

Data Preprocessing & Augmentation



Preprocessing



Resize

224 × 224 pixels



Normalization

Scale to [0, 1]



Class Weighting

Handle imbalance



Augmentation Techniques



Rotation

±15° random



Zoom

0.8-1.2× scale



Flipping

Horizontal & Vertical

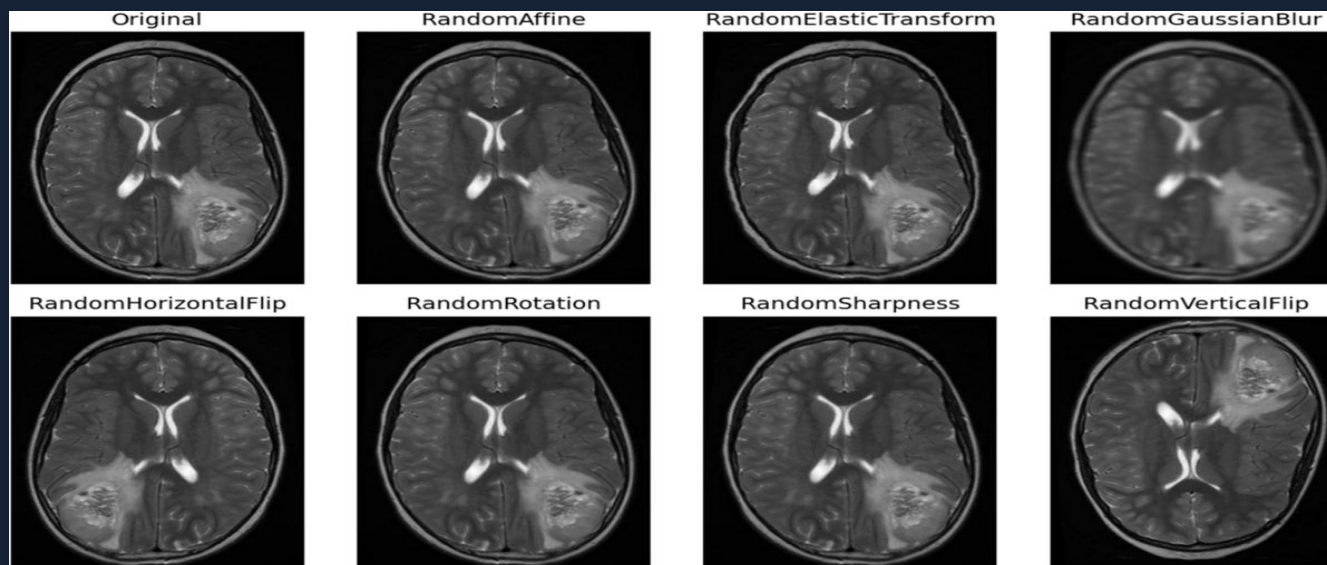


Brightness

±20% intensity



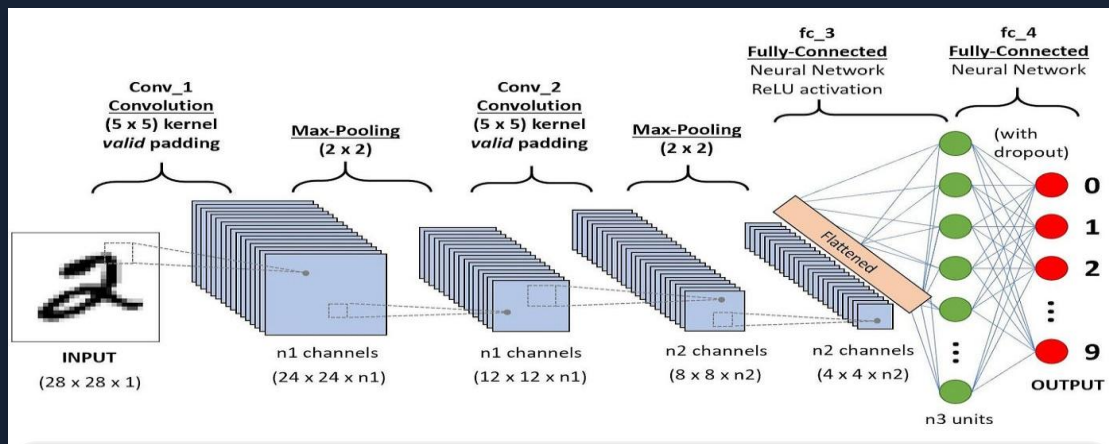
Augmentation Examples



Baseline CNN Architecture



Network Structure



Input Layer

224 × 224



Convolutional Layers

32 → 64 → 128 filters



ReLU Activation

Non-linearity injection



Max Pooling

2 × 2 downsampling



Performance Analysis

Test Accuracy

62%



Low



Poor

Generalization



High

Confusion



Limited

Features



Visual Similarity Issue

Glioma and Meningioma show high confusion due to similar appearance



Training from Scratch

Limited feature learning without pretrained weights

EfficientNetB0 Architecture

Compound Scaling



 **Depth**
Number of layers

 **Width**
Channels

 **Resolution**
Image size

MBConv Blocks

 **Depthwise Separable Convolutions**
Reduces FLOPs drastically without losing accuracy

 **Squeeze-and-Excitation (SE)**
Adaptive channel-wise feature recalibration

Performance

77.2%

Accuracy



Efficient



Fast

 High accuracy with fewer parameters

Balance achieved

ResNet50 Architecture

🏆 CHAMPION MODEL



Architecture



50 Layers Deep

Deep residual learning



Skip Connections

Identity shortcuts bypass layers



Bottleneck Blocks

1×1, 3×3, 1×1 convolutions

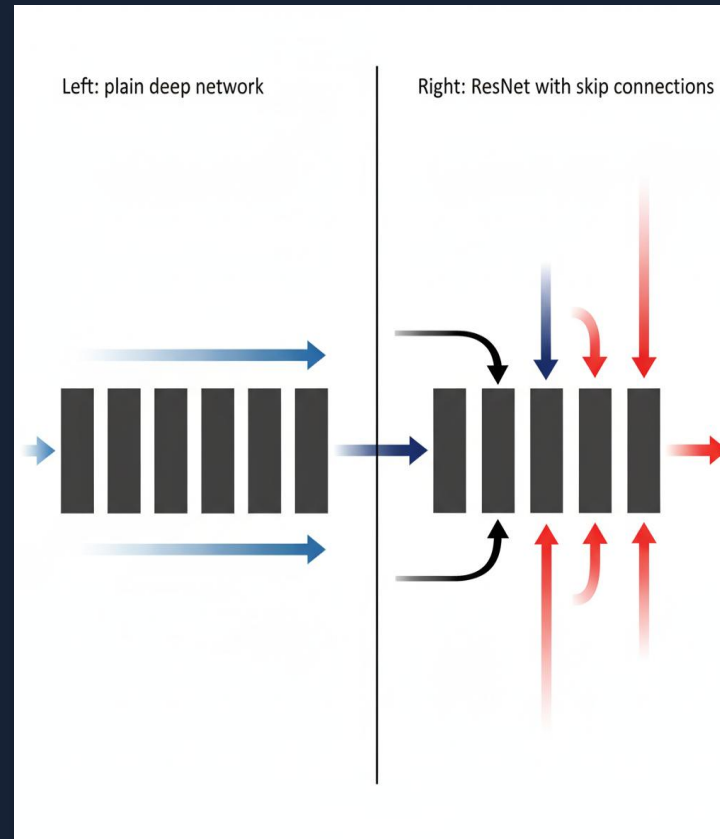


Solves Vanishing Gradients

Better gradient flow



ResNet Structure



Advantages

Test Accuracy

94.8%



Highest Performance



Skip Connections

Enables deep training



Transfer Learning

ImageNet pretrained weights



Complex Pattern Capture

Superior feature learning



Stable Training

Smooth loss curves

Training Strategy & Optimization



Optimization



Adam Optimizer 1×10^{-5}

Adaptive learning rate



Cross-Entropy Loss
Multi-class classification



Early Stopping
Prevent overfitting



LR Scheduling
Reduce on plateau



Fine-Tuning

1

Feature Transfer

Initialize with
ImageNet pretrained weights

2

Specialized Fine-Tuning

- Freeze lower layers
- Fine-tune deeper layers
- Low learning rate



Augmentation



Rotation
 $\pm 15^\circ$



Flipping
H / V



Zoom
0.8 - 1.2x



Brightness
 $\pm 20\%$



For All 4 Classes

Glioma, Meningioma,
Pituitary, No Tumor



Benefits



Faster Convergence

Pretrained features accelerate learning



Better Generalization

Balanced across all 4 tumor classes



Reduced Overfitting

Augmentation & early stopping



Validated on All Categories

4 classes tested and optimized

Evaluation Metrics



Accuracy

Overall correctness rate



Precision

True positive predictions



Recall

True positive detection rate
(Sensitivity)



F1-Score

Balance precision & recall



Clinical Importance



High Recall

Minimizes missed tumors. **Critical for patient safety** and early detection



High Precision

Reduces unnecessary treatments and false alarms



F1-Score Balance

Ensures balanced performance across both metrics



Confusion Matrix

		Actual Values	
		Positive	Negative
Predicted Values	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

TP

True Positive - Correct tumor detection

FP

False Positive - Wrong tumor prediction

FN

False Negative - Missed tumor detection



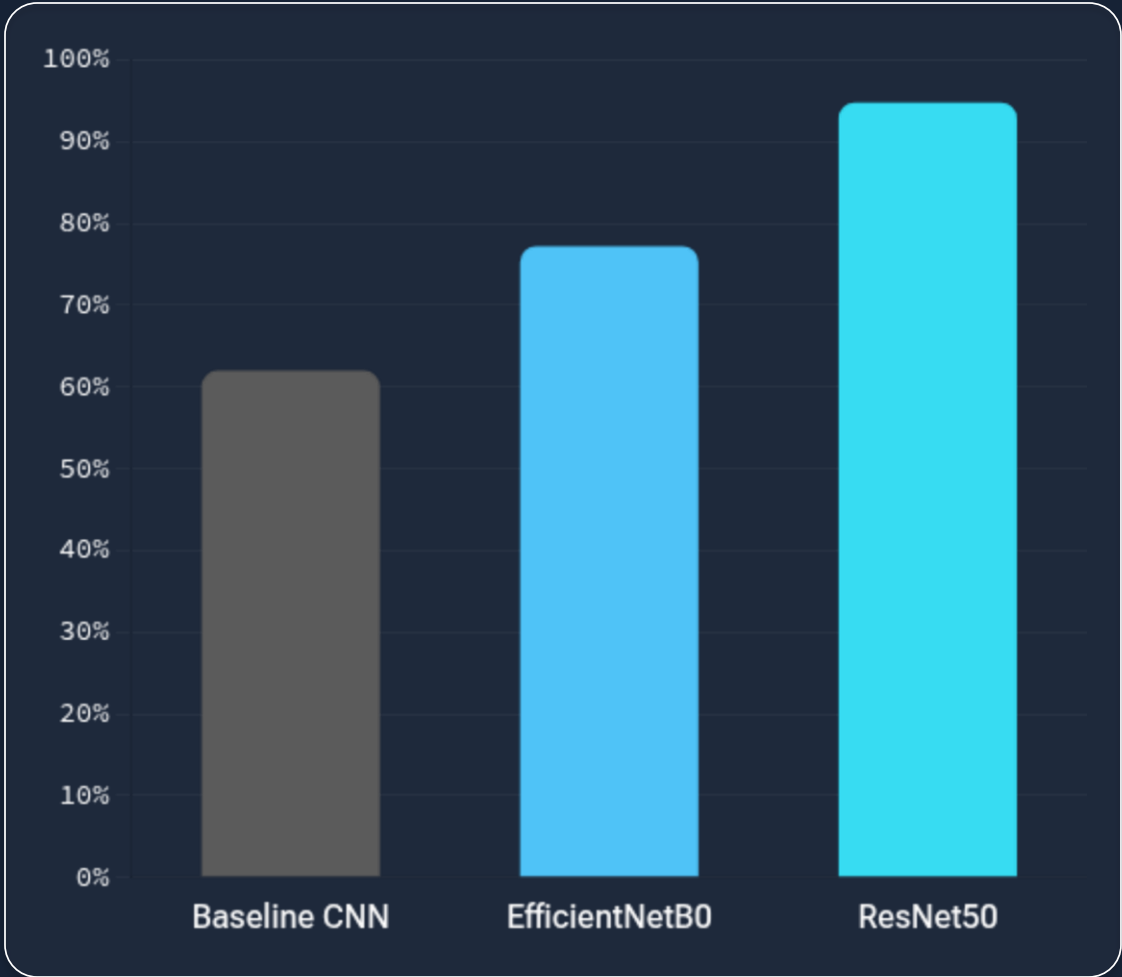
True Negative - Correct no-tumor detection

Results & Model Comparison



Model Accuracy

● Baseline CNN ● EfficientNetB0 ● ResNet50



Performance Highlights



ResNet50
94.8%

94.8%

Highest Accuracy



Skip Connections



Pretrained Weights



Best Performance



EfficientNetB0
77.2%



Good Accuracy



Efficient Design



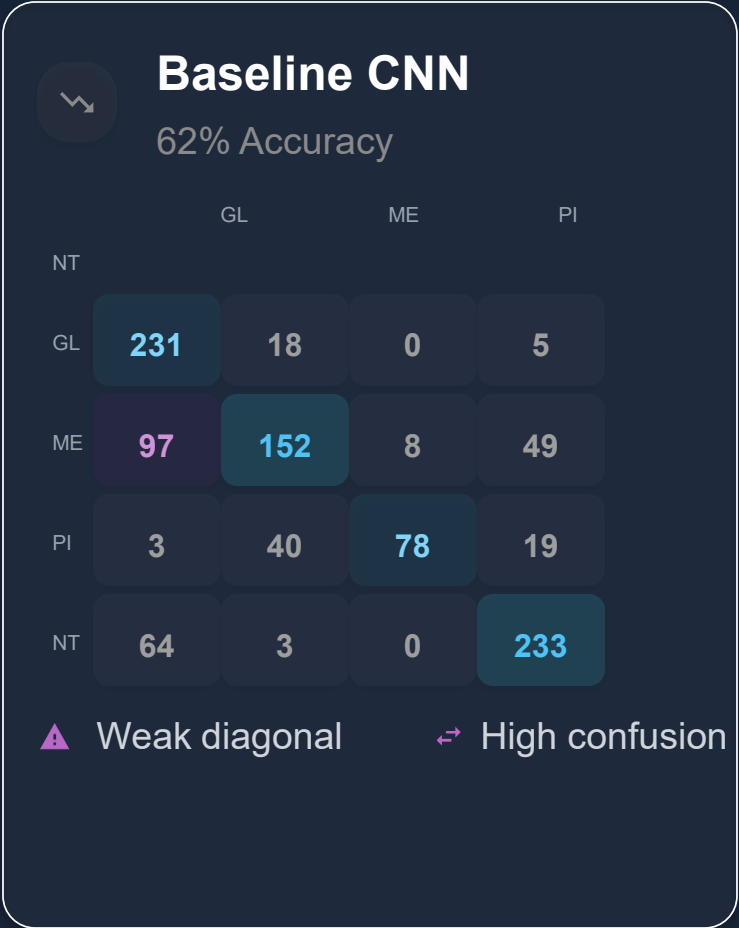
Baseline CNN
62%

Benchmark Only



High Confusion

Confusion Matrices Analysis



Key Insight

Higher diagonal values = better model performance

Visual Similarity

Glioma ↔ Meningioma show visual similarity confusion

Best Performance

ResNet50 achieves minimal off-diagonal values

Explainable AI with Grad-CAM

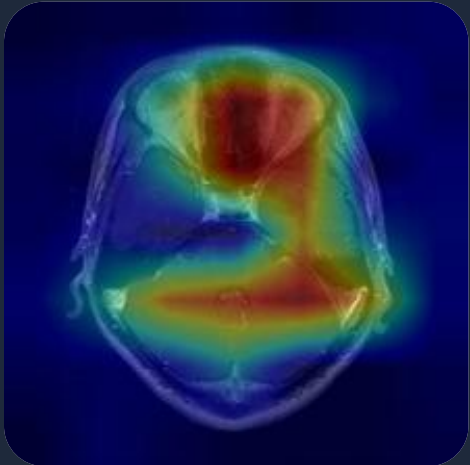


Gradient-weighted Class Activation Mapping

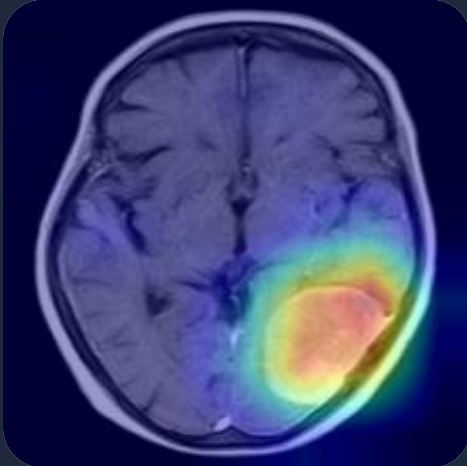
Visualizes which regions of the image **most influence** the model's prediction



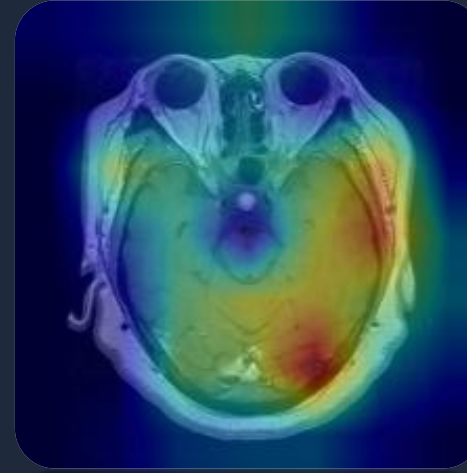
Glioma



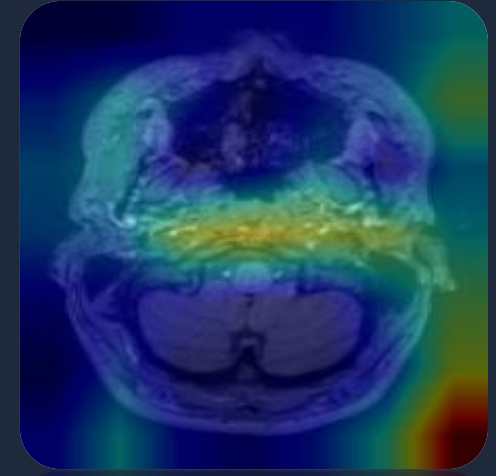
Meningioma



Pituitary



No Tumor



Clinical Importance



Builds trust by showing model focuses on **medically relevant regions**, not artifacts



Important Clarification

Grad-C**explanations**AM provides **qualitative**, not precise tumor segmentation

Deployment – Streamlit Application



Application Workflow

1

Upload MRI Image

User uploads brain MRI scan

2

Model Classification

AI identifies tumor type

3

Grad-CAM Heatmap

Highlights tumor regions

Technologies

Streamlit

TensorFlow

NumPy

PIL

Application Interface

Deploy

Settings 

Select AI Architecture

ResNet50

Upload MRI Scan

Drag and drop file here

Limit 200MB per file • JPG, JP...

Browse files

Grad-CAM Alpha

0.40

 **BRAIN AI DIAGNOSIS SYSTEM**

Advanced Clinical Decision Support Platform | v2.0

System Ready. Please upload a neuroimaging scan (MRI) via the sidebar to initiate analysis.

Developed by Brain AI Research Team | Decision Support Tool for Neuro-Oncology

Challenges & Limitations



Class Imbalance

Realistic clinical distribution requiring class weighting



Computational Cost

Deep models require significant GPU resources



Limited Data

Labeled medical imaging datasets are expensive



Interpretability

Black-box models require Grad-CAM explanations



Overfitting Risk

Small dataset sizes increase overfitting risk



Time Constraints

Academic project timeline limits development



Augmentation Limits

Cannot fully compensate for real-world variability



Domain Gap

Pretrained models on natural images have limitations



Mitigation Strategies



Class Weighting



Early Stopping



Transfer Learning



Grad-CAM

Conclusion & Future Work



Key Achievements



94.8% Accuracy

ResNet50 achieves highest performance



Grad-CAM Explainability

Visual explanations for clinical trust



Streamlit Deployment

Interactive demo application ready



Transfer Learning Effectiveness

- Baseline CNN: 62%
- EfficientNetB0: ~90%
- **ResNet50: 94.8%**

Pretrained models significantly outperform baseline

Future Directions



Larger Datasets

More diverse training samples



Real-time System

Clinical deployment pipeline



Multi-modal Integration

CT, PET imaging fusion



3D MRI Analysis

Volumetric feature extraction



Clinical Validation

Multi-center prospective studies



We thank Dr. Jamil Al-Sawwa for guidance and Tafila Technical University for resources



Thank You

Questions & Discussion